

Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

ANSWER KEY



Compound angle formula for cosine

- 1 Use the compound angle formula $\cos(A + B) = \cos A \cos B - \sin A \sin B$ to find the exact value of $\cos 75^\circ$, writing $75^\circ = 45^\circ + 30^\circ$.
- $\cos 75^\circ = \cos 45^\circ \cos 30^\circ - \sin 45^\circ \sin 30^\circ$. Since $\cos 45^\circ = \sin 45^\circ = \sqrt{2}/2$, $\cos 30^\circ = \sqrt{3}/2$, and $\sin 30^\circ = 1/2$, this gives $\cos 75^\circ = \frac{\sqrt{6} - \sqrt{2}}{4}$.
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Compound angle formula for sine

- 2 Use the compound angle formula $\sin(A - B) = \sin A \cos B - \cos A \sin B$ to find the exact value of $\sin 15^\circ$, writing $15^\circ = 45^\circ - 30^\circ$.
- $\sin 15^\circ = \sin 45^\circ \cos 30^\circ - \cos 45^\circ \sin 30^\circ$. Since $\sin 45^\circ = \cos 45^\circ = \sqrt{2}/2$, $\cos 30^\circ = \sqrt{3}/2$, and $\sin 30^\circ = 1/2$, this gives $\sin 15^\circ = \frac{\sqrt{6} - \sqrt{2}}{4}$. (This matches $\cos 75^\circ$ above, as expected since $\sin 15^\circ = \cos 75^\circ$.)
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Compound angle formula for tangent

- 3 The compound angle formula for tangent is $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$. Use it with $A = 45^\circ$ and $B = 30^\circ$ to find the exact value of $\tan 15^\circ$, giving your answer in simplified surd form.
- Substituting $\tan 45^\circ = 1$ and $\tan 30^\circ = 1/\sqrt{3}$: $\tan 15^\circ = (1 - 1/\sqrt{3})/(1 + 1/\sqrt{3})$. Multiplying top and bottom by $\sqrt{3}$: $\tan 15^\circ = (\sqrt{3} - 1)/(\sqrt{3} + 1)$. Rationalising by multiplying top and bottom by $(\sqrt{3} - 1)$: the numerator becomes $(\sqrt{3} - 1)^2 = 4 - 2\sqrt{3}$ and the denominator becomes $(\sqrt{3} + 1)(\sqrt{3} - 1) = 2$. So $\tan 15^\circ = (4 - 2\sqrt{3})/2 = 2 - \sqrt{3}$.
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Proving identities using the compound angle formulas

- 4 Prove that $\sin(A + B) + \sin(A - B) = 2\sin A \cos B$.
- Expand both: $\sin(A + B) = \sin A \cos B + \cos A \sin B$ and $\sin(A - B) = \sin A \cos B - \cos A \sin B$. Adding them, the $\cos A \sin B$ terms cancel: $\sin(A + B) + \sin(A - B) = 2\sin A \cos B$, as required.
- 5 Prove that $\cos(A + B)\cos(A - B) = \cos^2 A - \sin^2 B$.
- Expand: $\cos(A + B) = \cos A \cos B - \sin A \sin B$ and $\cos(A - B) = \cos A \cos B + \sin A \sin B$. Their product is a difference of squares: $\cos^2 A \cos^2 B - \sin^2 A \sin^2 B$. Using $\cos^2 B = 1 - \sin^2 B$: this equals $\cos^2 A(1 - \sin^2 B) - \sin^2 A \sin^2 B = \cos^2 A - \sin^2 B(\cos^2 A + \sin^2 A) = \cos^2 A - \sin^2 B$, as required.
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Solving a trigonometric equation using a compound angle formula

- 6 Solve $\sin(x + 30^\circ) = \cos x$ for $0^\circ \leq x \leq 360^\circ$.

Expand the left side: $\sin(x + 30^\circ) = \sin x \cos 30^\circ + \cos x \sin 30^\circ$. Substituting exact values:
 $\sin(x + 30^\circ) = \frac{\sqrt{3}}{2} \sin x + \frac{1}{2} \cos x$. Setting this equal to $\cos x$ and rearranging: $\frac{\sqrt{3}}{2} \sin x = \frac{1}{2} \cos x$.
Dividing both sides by $\cos x$ and by $1/2$ gives $\tan x = 1/\sqrt{3}$. Since this has period 180° : $x = 30^\circ$ or $x = 210^\circ$.

Evaluating reciprocal trigonometric functions from a given ratio

- 7 Given $\cos \theta = \frac{3}{5}$ with θ in Quadrant I, find the exact values of $\sec \theta$, $\csc \theta$, and $\cot \theta$.

In Quadrant I all ratios are positive. By the Pythagorean identity, $\sin \theta = \sqrt{1 - 9/25} = 4/5$. Then $\sec \theta = 1/\cos \theta = 5/3$, $\csc \theta = 1/\sin \theta = 5/4$, and $\cot \theta = \cos \theta / \sin \theta = 3/4$.

Reciprocal trigonometric functions in another quadrant

- 8 Given $\tan \theta = -2$ with θ in Quadrant II, find the exact values of $\sec \theta$, $\csc \theta$, and $\cot \theta$.

In Quadrant II, $\sin \theta > 0$ and $\cos \theta < 0$. Treating $\tan \theta = -2$ as opposite = 2, adjacent = -1, the reference triangle has hypotenuse $\sqrt{5}$. So $\sin \theta = 2/\sqrt{5}$ and $\cos \theta = -1/\sqrt{5}$. Then $\sec \theta = 1/\cos \theta = -\sqrt{5}$, $\csc \theta = 1/\sin \theta = \sqrt{5}/2$, and $\cot \theta = 1/\tan \theta = -1/2$.

Deriving $1 + \tan^2 \theta = \sec^2 \theta$

- 9 Starting from the Pythagorean identity $\sin^2 \theta + \cos^2 \theta = 1$, divide through by $\cos^2 \theta$ to derive $1 + \tan^2 \theta = \sec^2 \theta$.

Dividing every term by $\cos^2 \theta$: $\frac{\sin^2 \theta}{\cos^2 \theta} + \frac{\cos^2 \theta}{\cos^2 \theta} = \frac{1}{\cos^2 \theta}$, which is exactly $\tan^2 \theta + 1 = \sec^2 \theta$.

Deriving $1 + \cot^2 \theta = \csc^2 \theta$

- 10 Starting from the Pythagorean identity $\sin^2 \theta + \cos^2 \theta = 1$, divide through by $\sin^2 \theta$ to derive $1 + \cot^2 \theta = \csc^2 \theta$.

Dividing every term by $\sin^2 \theta$: $\frac{\sin^2 \theta}{\sin^2 \theta} + \frac{\cos^2 \theta}{\sin^2 \theta} = \frac{1}{\sin^2 \theta}$, which is exactly $1 + \cot^2 \theta = \csc^2 \theta$.

Simplifying an expression using reciprocal identities

- 11 Simplify $\sec \theta - \cos \theta$, expressing your answer in terms of $\sin \theta$ and $\tan \theta$.

$\sec \theta - \cos \theta = \frac{1}{\cos \theta} - \cos \theta = \frac{1 - \cos^2 \theta}{\cos \theta} = \frac{\sin^2 \theta}{\cos \theta} = \sin \theta \cdot \frac{\sin \theta}{\cos \theta} = \sin \theta \tan \theta$.

Mixed application: combining reciprocal identities across quadrants

12 Given $\sec \theta = 13/5$ with θ in Quadrant IV, find the exact value of $\sin \theta + \cot \theta$.

$\cos \theta = 5/13$. By the Pythagorean identity, $\sin^2 \theta = 1 - 25/169 = 144/169$. Since θ is in Quadrant IV, $\sin \theta = -12/13$. Then $\cot \theta = \cos \theta / \sin \theta = (5/13) / (-12/13) = -5/12$. Adding a common denominator of 156: $\sin \theta + \cot \theta = -144/156 - 65/156 = -209/156$.